
```

clc;
clear all;
close all;

syms t s

% 1. Define the CT signal
x = exp(2*t) * sin(3*t) * heaviside(t);

% 2. Plot the signal over a finite interval
fplot(x, [-2 10])
title('x(t) = e^{2t} sin(3t) u(t)')
xlabel('t')
ylabel('x(t)')
grid on

% 3. Compute Laplace Transform
y = laplace(x, t, s);

% 4. Display and pretty-print the Laplace Transform
disp('Laplace Transform Y(s):')
disp(y)
pretty(y)

% 5. Compute Inverse Laplace Transform
x1 = ilaplace(y, s, t);

% 6. Display and pretty-print the inverse
disp('Inverse Laplace Transform x(t):')
disp(x1)
pretty(x1)

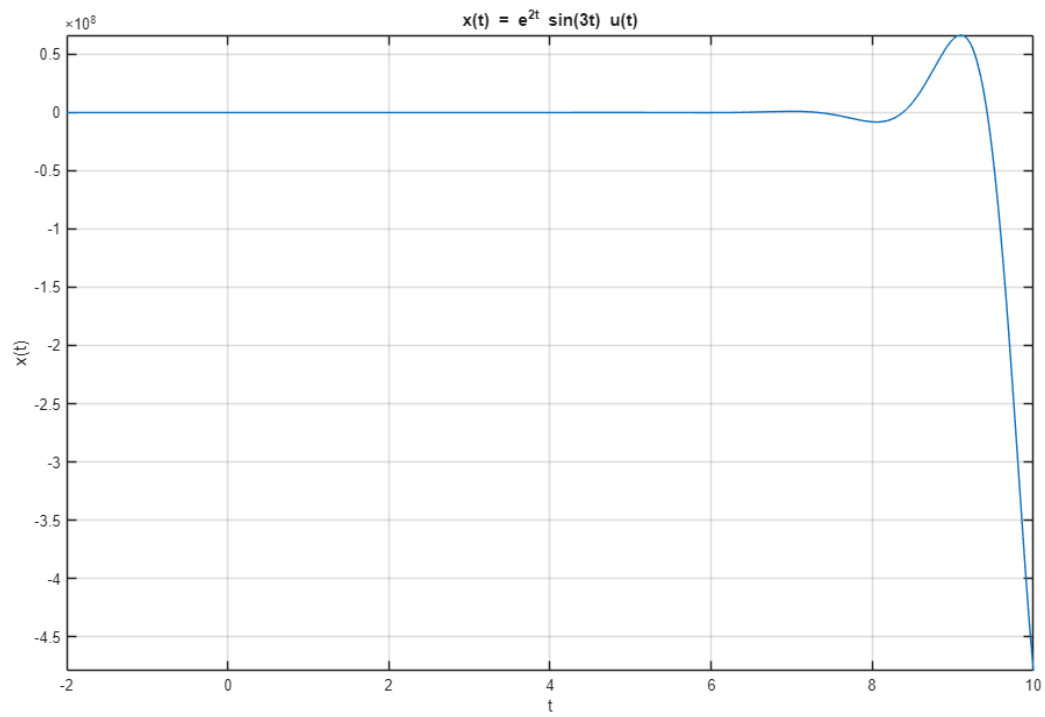
Laplace Transform Y(s):
3/((s - 2)^2 + 9)

      3
-----
      2
(s - 2)  + 9

Inverse Laplace Transform x(t):
sin(3*t)*exp(2*t)

sin(3 t) exp(2 t)

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